

Membership in the security domain of an operational group is determined by a Node's possession of all the epoch keys required to generate the current, valid *operational group key* for the group. Individual Nodes can be members of multiple operational groups simultaneously. The set of groups to which a Node belongs can change over time as dictated by application requirements and policies. Groups MAY be introduced or withdrawn over time as need arises.

#### 4.17.1.1. Operational Group Ids

Operational groups are identified uniquely within a Fabric by a [Group Identifier](#). Administrators SHALL assign Group Ids such that no two operational groups within a Fabric have the same Group ID. It is assumed a given Administrator has sufficient access to centralized knowledge, so as to allocate unique Group Ids under a given Fabric such that there are no collisions.

Multiple operational groups MAY share the same operational group key, and thus be used to create logical subgroups over a shared security domain. Operational groups which do not share related functionality, such as a lighting group and a sprinkler group, SHOULD NOT share the same operational key. As an example policy, a lighting application could have all lighting Nodes share a single group key, while organizing lighting subgroups for various rooms or spaces within the structure by assigning a different Group ID to each such subgroup.

#### 4.17.2. Operational Group Key Derivation

An *operational group key* is a symmetric key used as the Encryption Key during [Message Processing](#) for group communication. An *operational group key* is produced by applying a key derivation function with an *epoch key* and salt value as inputs as follows:

```
OperationalGroupKey =  
    Crypto_KDF  
    (  
        InputKey = Epoch Key,  
        Salt = CompressedFabricIdentifier,  
        Info = "GroupKey v1.0",  
        Length = CRYPTO_SYMMETRIC_KEY_LENGTH_BITS  
    )
```

where `[]` denotes a zero-length array.

The **Info** portion of the key derivation is specified in [Section 4.17.2.1, “Group Security Info”](#). The **Salt** portion of the key derivation is specified in [Section 4.3.2.2, “Compressed Fabric Identifier”](#).

For example, given:

- An Epoch Key value of: `23:5b:f7:e6:28:23:d3:58:dc:a4:ba:50:b1:53:5f:4b`
- A CompressedFabricIdentifier value of: `87:e1:b0:04:e2:35:a1:30`

After the above derivation following the definition of **Crypto\_KDF** in [Section 3.8, “Key Derivation Function \(KDF\)”](#), the resulting operational group key would be: